

Material Safety Data Sheet

1. Identification of the product and of the company

Product name: RAE-Sep Benzene and Halocarbon Tubes

Manufacture/supplier information:

RAE Systems Inc.
3775 North First Street
San Jose, CA 95134 USA
Phone: 408-952-8200
Fax: 408-952-8480

2. Chemical Identification of ingredients

CAS number	Component	Percent, %
Not available	Inert material and glass of the tube	> 84
7631-86-9	SiO ₂	> 8
112926-00-8	Matrix, silica gel	< 7
7664-93-9	Sulfuric acid, H ₂ SO ₄ , < 75mg	< 1.5
	Chromium compounds, CrO _x , <10 mg	< 0.2

Inert porous carrier material impregnated with sulfuric acid and chromium compounds sealed in a glass tube. There is no release of these chemicals from the glass tube in normal use.

This product may be regulated, have exposure limits or other information identified as following: sulfuric acid (7664-93-9).

This product is considered hazardous under OSHA 29 CFR 1910.1200

3. Hazards Identification

Primary hazard: Broken glass

The following hazards are possible under abnormal handling conditions

Primary routes of entry: Inhalation? Yes Skin? Yes. Ingestion? Yes

Health Hazards (Acute & Chronic):

Sulfuric acid: Strongly corrosive to skin, eyes and respiratory tract. Exposure symptoms include burning sensation, coughing, wheezing, shortness of breath, nausea and vomiting.

NEPA RATING: Health: 3, Fire: 0, Reactivity: 1, Special Hazard: -W-

Sulfuric Acid has been determined to be a human carcinogen by the following authorities:

- IARC: 2012, Monograph 54, 100F (Acid mists, strong inorganic)(Group 1 (carcinogenic to humans).

- NTP: Known to be a human carcinogen, first listed in the ninth report (2000).

Silica: Inhalation of amorphous silica can cause irritation to the lungs. Exposure to the eye can cause extreme irritation.

Chromium Compounds: Strongly corrosive to skin, eyes, and respiratory tract. Matrix: May be harmful if ingested in large quantity. Irritating to eyes and respiratory system if inhaled as dust. Prolonged inhalation of dust may cause fibrosis of the lung.

4. First aid measures (in the case of contact with the contents of a broken tube)

Eyes: Immediately flush eyes for at least 15 minutes, while holding eyelids open. Seek immediate medical attention.

Skin: Wash with copious amounts of water and in severe cases, seek medical attention.

Ingestion: Do not induce vomiting, seek immediate medical attention.

Inhalation: Remove person to fresh air and in severe cases, seek medical attention.

5. Fire fighting measures

All components are non-combustible except the end-plug fibers. all known extinguishers can be used.

All known Fire fighting measures including water spray, foam, dry powder, sand or carbon dioxide can be used.

6. Accidental release measures

Do not pick up broken glass with bare hands if the tube is broken. Contain any larger leaks using a plastic vessel. Cover with lime, sand or soda ash. Very small leaks may be diluted with plenty of water or water mixture. Neutralize washings with calcium carbonate or dilute caustic and dispose of them according with local, State and Federal regulations.

7. Handling and storage

Wear gloves and eye protection when opening tubes to protect against glass pieces and sharp edges. During normal operation, users are not exposed to the hazardous components. The potential for hazardous exposure exists after the tubes are opened or physically damaged by crushing or breaking. If the tubes are left open for extended periods in highly humid environments, the potential exists for water to be absorbed into the tube and leach sulfuric and chromic acids. There is a potential for respiratory exposure from dust if broken and dried. Under typical ambient conditions the hygroscopicity helps minimize the likelihood of dust formation.

Store unopened tubes in the protective boxes at 0-10°C to preserve shelf life.

Store used tubes in a sealable non-metal container to contain any spills or leaks.

8. Exposure control/personal protection

Avoid skin contact through use of personal protective equipment. Avoid inhalation or ingestion. Wear safety glasses and protective gloves.

Exposure Limits

Component	USA-OSHA PEL(TWA)	ACGIH (TWA)	NIOSH(TWA)
Sulfuric Acid (CAS 7664-93-9)	1mg/m3	0.2mg/m3	1 mg/m3
Silicon Dioxide (CAS 7631-86-9)	6mg/m3	10mg/m3	6mg/m3
Silica Gel (CAS 112926-00-8)	11.5mg/m3 (80/6.99%)	4mg/m3 (total)	6mg/m3
Chromium Compounds	0.5mg/m3	0.5 mg/m3	0.5 mg/m3

9. Physical/Chemical properties

Component Description	SiO₂ Colorless particles	Silica Gel Colorless particles	H₂SO₄ Colorless Aqueous solution	Cr compound powder
Melting point	n/a	n/a	n/a	980 °C
Boiling point Vapor	n/a	n/a	330 °C (100%)	n/a
Pressure Vapor	n/a	n/a	n/a	n/a
Density	n/a	n/a	n/a	n/a
Evaporation rate	n/a	n/a	n/a	n/a
Density	2.6 g/cc	n/a	1.29g/cc	2.7 g/cc
Water solubility	insoluble	insoluble	miscible	v. soluble

10. Stability and Reactivity

Stability : Stable under ambient temperatures and pressures.

Conditions to Avoid: Avoid heat sources and contact with water.

Incompatibilities: Not applicable

Hazardous polymerization: will not occur.

11. Toxicological information

Sulfuric acid: strongly corrosive to skin, eyes, and respiratory tract.

Exposure symptoms include burning sensation, coughing, wheezing, shortness, of breath, nausea and vomiting.

Inhalation LC50 Rat: 510 mg/m3/2H

Inhalation LC50 Mouse: 320 mg/m3/2H

Oral LD50 Rat 2140 mg/

Chromium compound: strongly corrosive to skin, eyes and respiratory tract.

Hazardous if ingested.

12. Ecological information

Do not allow to enter water supplies, and food chain.

13. Disposal considerations

Dispose of in accordance with local, State or Federal Regulations

14. Transport information

This product does not pose significant risk to health, safety or property. Breakage of tubes caused by drops, high pressure or bends should be avoided.

15. Regulatory information

This product contains <75 mg or <1.5% (w/w) sulfuric acid and <10 mg or <0.2% (w/w). Cr-compound impregnated into silica gel in sealed glass tube.

This product contains the following components that are listed as carcinogenic on the California Proposition 65 list: Sulfuric Acid

The following components are identified under the Canadian Hazardous Products Act Ingredient Disclosure list:

Item(English)	Item(French)	Component	CAS #	Minimum Concentration
1403.	(1488)	Silica, amorphous, fumed	7631-86-9	1%
1485.	(138)	Sulfuric acid	7664-93-9	1 %

16. Other information

For more detail contact the manufacturer at the address on page 1 for MSDS of individual components.